

Oumou Kaltoum Sall

Data Scientist & ML Engineer | Applied Mathematics & Data Optimization

s.sall@mundiapolis.ma | +212 6 65 68 27 34 | GitHub: KalsoumDS

PROFESSIONAL SUMMARY

Results-driven Data Scientist with a rigorous quantitative foundation in Applied Mathematics and hands-on experience deploying machine learning architectures. Specialized in translating complex industrial constraints into numerical optimization workflows, structured data pipelines, and scalable predictive models. Expert in Python and PyTorch, combining academic excellence with an entrepreneurial drive to deliver production-grade data solutions for strategic decision-making.

TECHNICAL SKILLS

- **Data Science & Machine Learning:** Statistical Modeling, Numerical Optimization, Predictive Analytics, Spatial Data Fusion, Time Series Forecasting, Stochastic Processes, Linear Algebra, Scikit-learn, PyTorch.
- **Data Engineering & Tools:** Data Pipelines Orchestration (ETL), Database Management (SQL), Python (Expert), C++, Git (Version Control), Docker, Linux, Cloud Operations.
- **Generative AI Advanced Frameworks:** LangChain, Autonomous Agents, Structured RAG Pipelines, Context Retrieval, LLM APIs.
- **Soft Skills:** Industrial Problem-Solving, Data Storytelling, Cross-Functional Team Collaboration, High Autonomy.

PROFESSIONAL EXPERIENCE

ABA Technology (Nextronic)

Casablanca, Morocco

R&D Data Scientist

February 2026 – Present

- Design, develop, and deploy advanced Deep Learning architectures and mathematical models for real-time predictive analytics.
- Prototype scalable systems and implement robust validation frameworks to assess technical feasibility under real-world industrial constraints.
- Build high-fidelity data processing pipelines, transitioning R&D prototypes into clean, production-grade enterprise code.

Freelance Data Scientist & ML Consultant

Remote

Independent Projects

August 2024 – January 2026

- Partnered directly with international clients to translate complex business requirements into custom statistical and machine learning frameworks.
- Built end-to-end automated data pipelines, including web scraping engines, advanced data cleaning, and relational database modeling.
- Delivered tailored data visualizations and interactive metrics to interpret algorithmic outputs for executive stakeholders.

Refocus Child

Senegal

Data Analyst (Volunteer)

January 2023 – June 2024

- Monitored data integrity, structured metadata frameworks, and optimized tracking databases to streamline operational reporting.

KEY PROJECTS

- **Industrial Data Fusion & Time Series Pipeline:** Engineered an end-to-end automated platform integrating multi-source spatial and time-series data. Applied numerical optimization and statistical learning to dynamically monitor process performance.
- **TabularAI – AutoML Framework:** Developed an innovative software using LangChain and Python to automate exploratory data analysis, successfully transforming raw tabular inputs into explainable AI (XAI) insights and executive presentations.

EDUCATION

M.Sc. in Data Science & Artificial Intelligence

Mundiapolis University of Casablanca

Graduation: October 2026

Casablanca, Morocco

- **Core Pillars:** Advanced Machine Learning, Optimization Algorithms, Information Theory, Computer Vision, MLOps.
- **Advanced Modules:** Numerical Analysis for AI, Probabilistic Methods, High-Performance Data Engineering.

B.Sc. in Applied Mathematics

Institut Supérieur de Management (ISM)

2021 – 2024

Dakar, Senegal

- **Core Pillars:** Numerical Analysis, Linear Programming, Stochastic Processes, Matrix Theory, and Mathematical Modeling.

ACADEMIC & PROFESSIONAL REFERENCES

Prof. Samar Mouchawrab

VP Academic Affairs, Mundiapolis University

Email: s.mouchawrab@mundiapolis.ma

Zakaria Abouhanifa

AI Engineer, Nextronic (ABA Technology)

Email: z.abouhanifa@nextronic.io